Title

Solution Requirements – SB Stocks

Functional Requirements

FR1 – User Registration

The system shall allow new users to create an account using username, email, password, and contact details.

FR2 – User Authentication

The system shall authenticate registered users using email and password.

FR3 – Dashboard

The system shall provide a personalized dashboard containing virtual balance and trading information.

FR4 – Stock Browsing

The system shall allow users to browse available stocks.

FR5 – Stock Search

The system shall allow users to search stocks using company name or stock symbol.

FR6 – Stock Details

The system shall display stock information and latest available prices.

FR7 – Historical Data

The system shall display historical stock price information.

FR8 – Portfolio Creation

Users shall be able to create multiple virtual portfolios.

FR9 – Portfolio Management

Users shall be able to view and manage their portfolios.

FR10 – Buy Stocks

Users shall be able to purchase stocks using virtual funds.

FR11 – Sell Stocks

Users shall be able to sell stocks available in their portfolios.

FR12 – Transaction Management

The system shall record buy and sell transactions.

FR13 – Order Management

The system shall maintain stock order records.

FR14 – Profit/Loss Analysis

The system shall calculate portfolio profit and loss.

FR15 – Strategy Analysis

The system shall allow users to analyze historical data using a basic trading strategy.

FR16 – Profile Management

Users shall be able to view and update permitted profile information.

FR17 – Admin Dashboard

Admins shall be able to monitor system information.

FR18 – User Monitoring

Admins shall be able to view registered users.

FR19 – Transaction Monitoring

Admins shall be able to view transactions.

FR20 – Order Monitoring

Admins shall be able to view stock orders.

FR21 – Logout

The system shall allow authenticated users to securely logout.

Non-Functional Requirements

Security

Passwords must be securely hashed.

Protected APIs must require valid authentication.

Performance

The system should provide API responses within reasonable time under normal usage.

Usability

The user interface should be intuitive and responsive.

Scalability

The MVC architecture should support future feature expansion.

Maintainability

Frontend and backend code should follow modular structures.

Reliability

Trading operations should validate balance and holdings before modifying data.

Compatibility

The application should work in modern web browsers.

□ Technology Stack

Title

Technology Stack – SB Stocks

Frontend Technologies

React

React is used to develop reusable user interface components and dynamic application pages.

Vite

Vite provides the frontend development environment and optimized build process.

JavaScript

JavaScript is used to implement frontend and backend application logic.

React Router

React Router manages navigation between application pages.

Axios

Axios is used to communicate with backend REST APIs.

Context API

React Context API manages authentication and shared application state.

Chart Library

A charting library is used to visualize historical stock price information and performance data.

Backend Technologies

Node.js

Node.js provides the server-side JavaScript runtime.

Express.js

Express.js is used to create REST API endpoints and backend middleware.

JWT

JSON Web Tokens are used for authenticated API access.

bcrypt/bcryptjs

Password hashing is used to protect user credentials.

Database

MongoDB

MongoDB stores application data using collections.

Mongoose

Mongoose defines schemas and provides database interaction with MongoDB.

Development Tools

- Visual Studio Code
- Git
- GitHub
- npm
- Nodemon
- Postman or API testing tools
- Web browser developer tools

Architecture

The application follows the Model-View-Controller architectural pattern.